

Response to the Editor

Carbon Footprint and Carbon-Aware Selection of Document-Oriented, Relational, and Hybrid Object-Relational Storage for Image-Based Time-Series Workloads in Green IoT

We thank the Editor for a constructive and precise assessment, and for the opportunity to revise. Every point has been addressed, and the manuscript is substantially stronger as a result. In particular, the request for stronger evidence of applicability prompted a new set of experiments against real payloads recorded from a face-recognition-based attendance system in operational use. These both confirmed the paper’s central conclusion and uncovered a quantitative correction we would otherwise have reported uncorrected. We are grateful for that.

Below, each of the Editor’s points is quoted, followed by our response and the location of the change. One point is answered with partial rebuttal, which we set out explicitly rather than leave implicit.

1. Positioning within the existing literature

“The discussion should better position the proposed framework within the existing scientific literature, demonstrating more convincingly how it advances beyond previous studies rather than primarily summarizing the experimental findings.”

Response. We accept this fully. On re-reading, the Discussion explained *mechanism* and then moved directly to threats to validity, without returning to the literature at all. Every citation in that section was to vendor or technical documentation rather than to scholarly work. The Editor’s characterisation was accurate.

We have added a new Discussion subsection, “**Positioning against prior work**”, which states what changes relative to prior studies rather than restating what we measured:

- **An accepted engine ordering is inverted.** Prior benchmarks on scalar time-series data report MongoDB ingesting faster than TimescaleDB. Our measurements agree with that direction at small payloads ($> 6\times$ faster at 360p) and then show it does not survive its own premise: the ordering inverts as the payload grows, and MongoDB becomes *infeasible* at 6K under the default bucketing of a native time-series collection, and unrunnable from 4K on a constrained host. What the scalar literature presents as a property of the engines is a property of the payload regime those benchmarks occupy. We are careful in the manuscript to call this a qualitative comparison rather than a replication, since those studies use different data, hardware, and metrics, and to scope the literature claim to the studies compared in Table 1.
- **The carbon-accounting literature supplies methods, not answers at this granularity.** Green Algorithms, the Software Carbon Intensity specification, and the embodied-carbon literature can attribute a footprint to a workload once its energy is known, but cannot indicate which architecture will consume that energy. We measure per architecture, per operation, and per resolution, and we note that a single aggregate figure per architecture, which those methods would ordinarily yield, would have averaged the crossover away and produced a spurious universal winner.
- **The output is a decision instrument, not a benchmark table.** The framework is now expressed as a transferable calibration procedure (see point 3) rather than a recommendation

tied to our hardware. We are careful to claim that the procedure and the sequence of regimes transfer, not the payload sizes at which our own boundaries fall.

We also state explicitly what we do *not* claim: that the specific crossover payloads generalise. Table 1 has additionally been extended with a column recording what each prior study concludes and what changes in our payload regime, so the advance is visible in the comparison itself rather than asserted in prose.

2. Dialogue with recent publications in this journal

“The manuscript would also benefit from a stronger dialogue with recent publications in *Environment Systems and Decisions*, reinforcing its contribution to the journal’s body of knowledge.”

Response. Accepted. The submitted version cited no work from this journal. We have added a new Related Work subsection engaging six articles from it substantively, spanning 2019 to 2025. Each is used to sharpen a specific methodological choice, not cited in passing:

- A systematic review of multi-criteria decision making for material selection [1] identifies a recurring weakness in decision frameworks: criteria are often taken from the literature rather than measured in the adopting organisation’s own context. This directly motivates our choice to key the rule to a variable the adopter can measure, namely payload size, and to supply a procedure for locating the boundaries locally rather than a fixed recommendation.
- A life-cycle assessment of lightweight railway vehicles [2] establishes sensitivity analysis as what separates a robust conclusion from an artefact of assumed parameters. We identify our multi-region grid-intensity analysis as playing exactly that role, and show the ranking is invariant to the parameter that varies most between deployment locations, national grid intensity differing by more than an order of magnitude across the regions evaluated.
- A data-driven study of smart-home energy consumption [3] treats building instrumentation as the object of environmental analysis. We draw the distinction that such models account for the energy the building consumes, but not for the tier that stores what the instrumentation records, which we show can be reduced by roughly half through a storage decision alone.
- Two studies of IoT operational integrity, on self-healing through log analytics [4] and real-time risk at the edge [5], both presuppose a data tier that durably retains device output. We supply the missing environmental cost of that presupposition.
- A review of price-driven energy management and demand response [6] prompts a substantive point rather than a citation: because emissions are energy multiplied by grid intensity, temporal or geographic load shifting rescales every architecture equally and cannot change which one to select. The saving must come from the architecture itself.

3. Practical and scientific value, and applicability beyond our environment

“At present, the framework is derived almost entirely from the experimental environment developed by the authors, and its applicability beyond the evaluated hardware, software configuration, and workload characteristics remains insufficiently discussed.”

Response. This was the most substantive point, and we have addressed it in three ways: analytically, empirically with new experiments, and by separating claims that depend on measurement from those that do not. We also offer one partial rebuttal, stated at the end.

3a. The mechanism behind the crossover, and how to locate it elsewhere

We now set out explicitly why the crossover exists. Each architecture pays a fixed cost per operation, independent of payload, and a payload-dependent cost that grows with the bytes written. MongoDB has the smallest fixed cost and the steepest growth, the hybrid is its mirror image, and inline PostgreSQL lies between on both counts, which is why it is best at no single operation and worst at none. An architecture with the lowest fixed cost therefore wins at small payloads and one with the shallowest growth wins at large ones, and because the orderings differ, the curves cross.

We deliberately stop short of a closed-form law. An earlier draft of this revision fitted a linear model and reported its constants; we discarded that, because the growth is not linear. MongoDB’s per-frame cost rises by more than two orders of magnitude across the measured range and accelerates sharply once payloads exceed roughly one megabyte, and fitting a straight line across that range produces a negative intercept, which is unphysical. What the measurements support is the ordering of the regimes and the interval in which each transition falls, and that is what we now report: crossovers are given as brackets bounded by adjacent measurement points, with no interpolation implied.

The consequence for transferability is a procedure rather than a formula, and the manuscript now states it as a numbered protocol so that it can be followed rather than inferred: measure the payload the cameras actually produce; fix the operation profile; bracket each candidate architecture at three or four payload sizes spanning that range; compare with run-to-run uncertainty, treating any separation inside it as a tie; exclude cells an architecture cannot complete; and select. It requires no model and no fitting, and it yields the boundary for the adopter’s installation rather than ours. We are explicit that what transfers is this procedure and the sequence of regimes it recovers, not the payload sizes at which our own boundaries fall.

3b. New experiments: a second machine, a real-case corpus, and what each isolates

We report three sets of new measurements, described in the manuscript as part of the study rather than as additions. We are explicit in the manuscript about what each does and does not isolate: none of them varies a single factor cleanly, and we state the confounds rather than leaving them to be inferred.

The same payloads on a second machine. We repeated the entire controlled sweep on a markedly smaller computer: four cores and 15.30 GB against eight cores and 30.99 GB, with the database cache scaled to the same proportion of host memory. Payload sizes, database images and benchmark protocol are held constant; the operating-system point release, container runtime, Python patch level and storage device differ alongside the processor and memory, so we attribute the difference to the environment rather than to the processor alone, and say so in the manuscript. Two results follow. The ordering at small payloads is unchanged, so that conclusion does not depend on the machine. The crossover, however, moves: the hybrid overtakes MongoDB one resolution step earlier on the smaller host, and at 1.40 MB MongoDB emits $0.41\times$ the hybrid’s carbon on the larger machine but $1.01\times$ on the smaller. We do not read the latter as establishing a crossover at that payload: a 1% separation sits well inside that host’s run-to-run variability, which reaches 28%, and each carbon value is a single integrated measurement. What the pair does establish is that a margin of more than a factor of two on one machine has disappeared on the other at the same payload, which is larger than the variability. This is the direction the

mechanism predicts, since the hybrid’s benefit is keeping bytes out of the database write path and a smaller host is where that path is most constrained.

A second infeasibility, of a different kind. On the smaller machine MongoDB ceased to be runnable from 4K upward. Its ingest rate fell from 3,973 rows per second at 360p to between three and five at 4K, and the kernel terminated the process under a global memory constraint; inline PostgreSQL and the hybrid completed every resolution to 7.13 MB on the same host in the same session. The cause follows from the architecture: MongoDB holds documents uncompressed in cache, so one insertion run at 4K is 5.8 GB of live working set that quality-90 JPEG gives it no way to shrink, whereas PostgreSQL relocates the payload out of line and the hybrid never admits it to the database. We report this as a finding rather than an incident. The 16 MiB document ceiling already in the paper is a protocol limit that no hardware removes; what depends on configuration is whether several frames are packed into one document that breaches it, and under the default bucketing we tested they are. This second limit is instead a resource limit that a modest host imposes, and a practitioner will meet it first. We also state the two qualifications: the cache held the same proportion of memory as on the larger machine, and neither host had swap.

A real-case corpus, and what it corrects. The submitted benchmark stored one image in every row, which is exactly reproducible but not what a camera produces. We recorded frames from a face-recognition-based attendance system in operational use, one networked and two USB cameras, and re-ran the benchmark with a different real image in every row. The architectural conclusion holds on two cameras, under a profile counting one insertion, one bulk retrieval and one latest-frame read equally. We are careful about what this establishes: the real corpus was measured on the second machine only, so payload and environment vary jointly between this experiment and the two-host comparison, and the manuscript claims consistency across the tested host and corpus combinations rather than hardware independence.

The correction itself is the more valuable result. Storing one repeated image rather than distinct ones understates MongoDB’s ingestion carbon by 2.4 to $2.9\times$ and its storage amplification by 16 to $25\times$. All three architectures change, but MongoDB alone changes for the worse; inline PostgreSQL and the hybrid become 19 – 25% and 39 – 47% *cheaper* with distinct frames, which we report rather than dismiss as noise, noting that the compared payloads are matched only approximately and that size and distinctness are not fully separable in the pairing. What distinguishes MongoDB is the sign, and the cause is that it alone compresses *across* rows. We report this prominently and against our own interest: the direction of our published result is confirmed and its magnitude is not.

Taken together these give a result we consider more useful than the resolution bands we originally proposed. Resolution is a label for payload size, not the deciding variable, and the substitution fails in practice: at 1080p on one machine, and under an ingestion-only profile, our collage payload selects MongoDB and real camera output selects inline PostgreSQL. Same resolution, same host, same software, opposite recommendation. The framework is therefore now stated in measured payload size, with resolution as an indicative guide whose reliability is specific to the camera.

3c. Separating structural claims from measured ones

Two claims are now identified as independent of our hardware entirely: the 16 MiB BSON ceiling is a protocol limit, so under the default bucketing we tested no hardware makes time-series collections able to store 6K frames; and the ordering of per-byte costs follows from where each architecture places the payload, not from how fast the host runs. We separate these deliberately from the measured boundaries, and we distinguish the immutable document ceiling from the bucket-formation failure we observed, which depends on how the collection is configured.

3d. Partial rebuttal: the single-host baseline is a design choice

Respectfully, we retain the single-host, single-client design and wish to justify rather than silently amend it. Multi-client concurrency changes how energy is attributed and introduces contention noise. It is effectively a different study, not a more realistic version of this one. The controlled baseline isolates payload handling, the variable under study, from scheduling and contention confounds, and keeps the comparison fully reproducible. We have strengthened the evidence for generality by the means above, in particular by repeating the entire sweep on a second machine, which we believe answers the Editor’s concern more directly than added concurrency would: it varies the computing environment, which was one of the three dimensions raised, while leaving the measurement interpretable. We note there too that the second environment differs in more than its processor. A multi-client evaluation remains identified as future work.

4. Limitations in the abstract

“The abstract should also explicitly acknowledge the main limitations of the research.”

Response. Accepted and implemented. The abstract now states, in the authors’ own words, that the study is deliberately bounded, and names its main limitations: the single-host, single-client controlled baseline rather than a distributed multi-tenant deployment; the measurement boundary, which covers the storage tier alone and excludes the deployment’s capture and face-recognition pipeline; a limited photographic corpus, with real-frame validation reaching 1080p; a single integrated carbon measurement per cell, with run-to-run variability reaching 28% on the smaller host; measurement under one national grid intensity, with the explicit note that the *ranking* is grid-invariant because emissions scale linearly with that factor; the annual deployment figures being a scenario projection for that workload rather than a saving measured against the installation as it currently runs; and embodied-carbon and fleet-scale figures being first-order estimates from published factors rather than site-specific measurements.

5. Declaration of AI-assisted technologies

“If artificial intelligence tools were used for language polishing or translation, this should be transparently declared in accordance with the journal’s editorial policies.”

Response. Declared. A **Use of AI-assisted technologies** statement has been added to Statements and Declarations, naming the three assistants used (Google Gemini, Anthropic Claude, and OpenAI Codex) and recording that they were used *solely for language editing*, that is grammar, clarity, and phrasing of author-written text, and that no AI tool was used to generate, analyse, or interpret the experimental data, produce figures or tables, or draw the scientific conclusions. The authors reviewed and edited all such output and take full responsibility for the entire content. AI is not credited as an author.

We took the opportunity to complete the declarations block, which was otherwise thin: Competing Interests, Data and Code Availability, and Ethics have also been added. The reproducibility artifact has been rebuilt for this revision so that it matches what the availability statement claims: it now carries the measurements from both hosts and all three cameras, the recorder, the figure-generation scripts for the new figures, and the full corpus of 8,724 recorded frames, released as one archive per camera. The ethics statement states precisely what was done and is careful not to claim more than the evidence supports. It records that no recognition function was invoked and no enrolled identity or template was accessed; that recording was carried out with no other person present and no biometric template computed or stored; that every saved frame

was passed through a named face detector at a stated threshold and none produced a detection; and that this is evidence the corpus contains *no detectable faces*, not a proof that it contains no personal data, since a detector sweep cannot establish the latter. It states the basis on which the work was treated as exempt rather than asserting an approval that was not sought.

Summary of changes

Editor’s point	Where addressed
1. Positioning in the literature	New Discussion subsection “Positioning against prior work”; Table 1 extended with a column stating what each prior study concludes and what changes here.
2. Dialogue with this journal	New Related Work subsection engaging six articles from this journal, spanning 2019–2025, each used to sharpen a specific methodological choice.
3. Applicability beyond our setup	New subsection deriving the mechanism behind the crossover and giving a numbered calibration protocol for locating it elsewhere; the entire sweep repeated on a second machine and reported side by side; a real-case corpus from an operational deployment; a second infeasibility arising from memory rather than protocol; and the framework restated in measured payload size after resolution proved an unreliable proxy.
4. Limitations in the abstract	Abstract revised to name the study’s main limitations explicitly, including the storage-tier measurement boundary and the scenario-projection status of the annual deployment figures.
5. AI declaration	Statements and Declarations rewritten, adding use of AI-assisted technologies (with the tools named), competing interests, data and code availability, and an ethics statement.

References

All six appear in *Environment Systems and Decisions*. Every DOI was verified to resolve and the bibliographic details checked against Crossref before inclusion.

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2. Kraft S (2023) Life cycle assessment in the development process of lightweight railway vehicles using sensitivity analysis. *Environment Systems and Decisions* 44(3):527–546. 10.1007/s10669-023-09955-6
3. Adams S, Greenspan S, Velez-Rojas M, Mankovski S, Beling PA (2019) Data-driven simulation for energy consumption estimation in a smart home. *Environment Systems and Decisions* 39(3):281–294. 10.1007/s10669-019-09727-1
4. Singh P, Saman Azari M, Vitale F, Flammini F, Mazzocca N, Caporuscio M (2022) Using log analytics and process mining to enable self-healing in the Internet of Things. *Environment Systems and Decisions* 42(2):234–250. 10.1007/s10669-022-09859-x

5. Radanliev P, De Roure D, Van Kleek M, Ani U, Burnap P, Anthi E (2020) Dynamic real-time risk analytics of uncontrollable states in complex internet of things systems: cyber risk at the edge. *Environment Systems and Decisions* 41(2):236–247. 10.1007/s10669-020-09792-x
6. Al Essa MJM (2025) A review on price-driven energy management systems and demand response programs in smart grids. *Environment Systems and Decisions* 45(1). 10.1007/s10669-024-09998-

We hope the revised manuscript now meets the standard required to proceed to an Associate Editor, and we thank the Editor again for guidance that materially improved the work.